Report:

Student #: 300310115

One reason the AG News Dataset got picked? It's been around a while, also called AG's News Corpus, so researchers trust it. Since it shows up often in NLP work, especially for sorting text, using it here feels solid, nothing forced. Articles come from more than 2000 outlets, split across four main themes: World, Sports, Business, then Science and Tech. Because labels are clean and layout is consistent, spotting patterns doesn't take heroic effort. Realistic samples sit inside, close to what actual news looks like online. Working through exploration or checking data health goes smoothly, mainly because heavy cleanup isn't needed before starting. No niche expertise stands in the way either.

Accessing the data happened through the Hugging Face Datasets Library; this tool makes loading big public collections consistent and repeatable. Though the AG News training portion holds around 120,000 entries, that volume isn't always useful when exploring patterns, shape, or cleanliness of information. Instead of working with everything, just 10,000 examples got pulled at random to keep things fast without losing real-world reflection. Before picking those cases, the entire set was shuffled using one unchanging seed number so outcomes stay fair and others can match them later.

Loaded up, it turned into a pandas DataFrame for easier look-through and visuals. This sample holds ten thousand entries, each with two main parts: one piece being the article text, also including a number tag showing its group. String format keeps the words intact, whereas integers handle labelseach between zero and three for those four groups. Nothing else tagged along here; just these pieces made it in, cutting down extra noise during review.

Starting off, we looked at the data to get a feel for how it spreads out and what stands out. Turns out, the labels are fairly even across categories no big tilt toward one side. Instead of lumping everything together, checking article lengths showed they differ quite a bit; most sit around a normal size, though some trail far below or stretch way above. On top of that, scanning for blanks turned up nothing missing in both text and category tags, which means the raw material came through clean.

For checking how good the data is, we brought in the Great Expectations tool during analysis. Starting off, a working environment from Great Expectations got set up; after that, the table made in pandas became part of it as usable information. Rules were added one by onemaking sure key columns stay full, class tags sit inside correct numbers, while written parts must act like proper words. When tested, everything passed without big issues showing up anywhere. Only thing standing out: some lines of text came way too brief, possibly needing cleanup later when building models.

Looking at things loosely, the AG News collection seems tidy, neatly arranged, right for training models. Because categories spread evenly, nothing's blank, labels stay steady, it fits supervised methods just fine. What comes through in this early look hints that next steps lean more toward how words get split, shortened, managed when too longless about scrubbing messy info.

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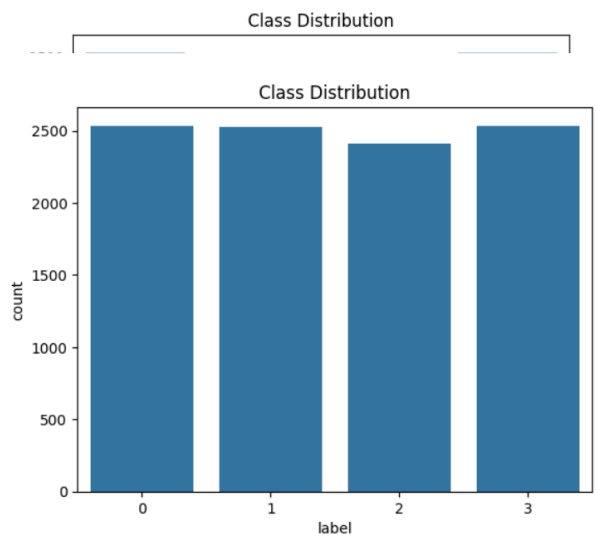
Appendix

[26]:

	text	label
count	10000	10000.000000
unique	10000	NaN
top	Bangladesh paralysed by strikes Opposition act...	NaN
freq	1	NaN
mean	NaN	1.494700
std	NaN	1.123876
min	NaN	0.000000
25%	NaN	0.000000
50%	NaN	1.000000
75%	NaN	3.000000
max	NaN	3.000000

[27]:

```
sns.countplot(x="label", data=df)
plt.title("Class Distribution")
plt.show()
```



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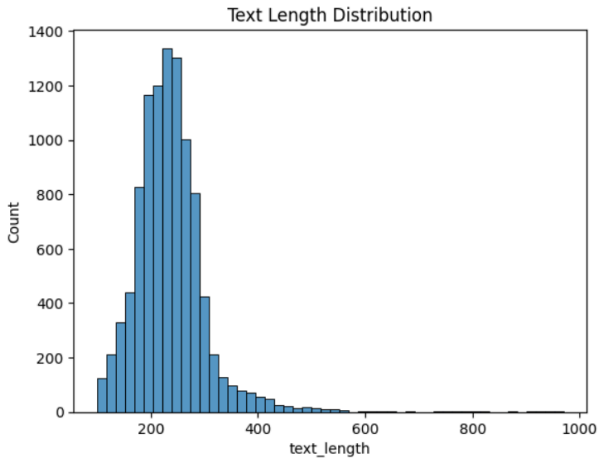
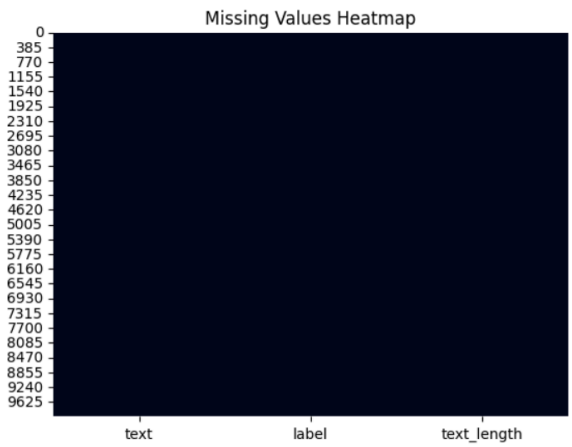
[28]:
```
df["text_length"] = df["text"].apply(len)

sns.histplot(df["text_length"], bins=50)
plt.title("Text Length Distribution")
plt.show()
```



[29]:
```
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Values Heatmap")
plt.show()
```



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```
c:\users\pc\eda_env\lib\site-packages\great_expectations\expectations\expectation.py:1404: UserWarning: `result_format` configured at the Validator-lev
el will not be persisted. Please add the configuration to your Checkpoint config or checkpoint_run() method instead.
  warnings.warn(
Calculating Metrics: 100%|██████████| 6/6 [00:00<00:00, 139.51it/s]
c:\users\pc\eda_env\lib\site-packages\great_expectations\expectations\expectation.py:1404: UserWarning: `result_format` configured at the Validator-lev
el will not be persisted. Please add the configuration to your Checkpoint config or checkpoint_run() method instead.
  warnings.warn(
Calculating Metrics: 100%|██████████| 6/6 [00:00<00:00, 117.66it/s]
c:\users\pc\eda_env\lib\site-packages\great_expectations\expectations\expectation.py:1404: UserWarning: `result_format` configured at the Validator-lev
el will not be persisted. Please add the configuration to your Checkpoint config or checkpoint_run() method instead.
  warnings.warn(
Calculating Metrics: 100%|██████████| 8/8 [00:00<00:00, 133.34it/s]
c:\users\pc\eda_env\lib\site-packages\great_expectations\expectations\expectation.py:1404: UserWarning: `result_format` configured at the Validator-lev
el will not be persisted. Please add the configuration to your Checkpoint config or checkpoint_run() method instead.
  warnings.warn(
Calculating Metrics: 100%|██████████| 1/1 [00:00<00:00, 71.25it/s]
Calculating Metrics: 100%|██████████| 13/13 [00:00<00:00, 181.76it/s]
```


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          "column": "text"
```

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```

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      "unexpected_count": 0,
      "unexpected_percent": 0.0,
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```

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```
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      "exception_traceback": null,
      "exception_message": null
    }
```



```
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```